

When does distribution shift break graph neural networks calibration?

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Abstract

Graph neural networks (GNNs) are increasingly deployed in real-world applications where distribution shift is unavoidable. However, how such shifts affect model calibration, defined as the agreement between predictive confidence and actual accuracy, remains poorly understood, and existing graph calibration methods typically rely on labeled validation data from the deployment distribution. In this work, I present the first closed-form theoretical characterization of GNN calibration under distribution shift. I show that calibration is governed by a single scalar quantity that explicitly depends on structural changes between the source and target graphs, as well as feature quality. This characterization precisely identifies when a model becomes over-confident, under-confident, or remains calibrated, and directly yields the optimal temperature scaling strategy. I further extend the analysis to graph convolutional networks with symmetric normalization, multi-class classification, and covariate shift, and derive a theoretical upper bound on the expected calibration error. My analysis also reveals that, under homogeneous distribution shift, a single global temperature is theoretically optimal, providing a principled explanation for why more complex node-wise recalibration methods offer no additional benefit. Building on these theoretical insights, I propose STAC, a source-free, label-free calibration method. Experiments on synthetic benchmarks demonstrate substantial calibration improvements, while evaluations on five real-world graph datasets show that reliable calibration without target labels remains challenging despite the strong predictive power of the theory. These findings identify label-free accuracy estimation under distribution shift as the central unresolved challenge for practical calibration of GNNs and establish a theoretical foundation for future research on trustworthy graph learning. Project details are available at <https://github.com/AraoufBh/GNN-Calibration>.

Keywords: graph neural networks, calibration, distribution shift, homophily, temperature scaling, uncertainty quantification

1. Introduction

Graph neural networks (GNNs) [1] are now routinely deployed in settings where the graph seen at test time differs from the one used for training: social and financial networks evolve, sensor and infrastructure graphs are rewired after failures or upgrades, and citation, molecular or recommendation graphs used at inference are drawn from different sub-populations than those used to train the model. In many of these settings raw predictive accuracy is not the primary bottleneck; what determines whether a GNN can be trusted inside a fraud-review, clinical-triage, or infrastructure-monitoring pipeline is whether its confidence scores are *calibrated*: whether a prediction reported with 90% confidence is, in fact, correct 90% of the time. Miscalibration is not cosmetic: it silently corrupts every downstream mechanism that consumes probability outputs, from selective prediction and human-in-the-loop triage to risk-weighted decision rules and model ensembling.

Two observations make GNN calibration under distribution shift both practically important and theoretically awkward. (i) GNNs are miscalibrated in structure-dependent ways; unlike image classifiers (typically over-confident), they are often *under*-confident in-distribution, and the size of the gap correlates with graph properties such as homophily and node degree [2, 3]. (ii) Standard calibrators need labels from the test distribution: temperature scaling [4] and its graph-specific variants [2, 3, 5–7] all fit a temperature (or a temperature-valued function of the node) by minimizing negative log-likelihood on a labeled validation set drawn from the *same* distribution as training. Under shift that temperature is simply wrong, and no target labels are available to refit it.

Despite an increasingly rich empirical literature on GNN calibration (Section 2), no existing account explains, from first principles, *how* and *why* shift moves a GNN’s confidence away from its accuracy, *in which direction*, and *by how much*. Existing post-hoc graph calibrators (CaGCN, GATS, GETS, WATS, SimCalib) are all validated in-distribution and are, by construction, vulnerable to exactly the failure mode that matters most in deployment: a validation set that no longer represents the test distribution. Empirical studies of GNN reliability under shift [8] document that the problem is real and that architectural expressivity alone does not fix it, but stop short of a

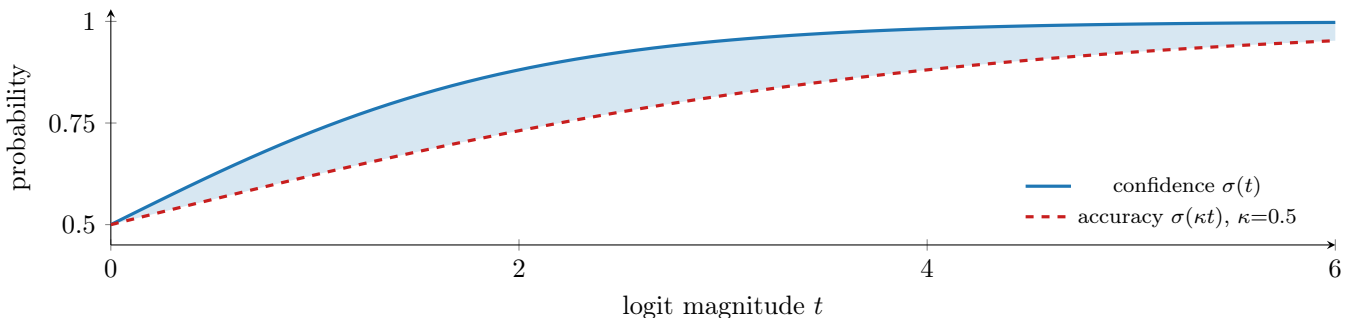


Figure 1: **The calibration slope.** A source-calibrated linear GNN reports confidence $\sigma(t)$ (blue) but its true accuracy is $\sigma(\kappa t)$ (red). The shaded area is the calibration error. When homophily/covariate shift makes $\kappa < 1$ the model is over-confident; a single temperature $T^* = 1/\kappa$ (i.e. $\sigma(t/T^*) = \sigma(\kappa t)$) removes the gap exactly.

mechanism. I supply that mechanism for a model simple enough to analyze in closed form yet expressive enough to reproduce, quantitatively, the calibration behaviour of trained GCNs in simulation and on five real graphs spanning two orders of magnitude in edge homophily (Fig. 1 previews the core result).

Contributions.. (C1) **A closed-form calibration slope** (Section 4): $\text{conf}(t) = \sigma(t)$ but $\text{acc}(t) = \sigma(\kappa t)$, with κ explicit in homophily and signal-to-noise ratio (SNR); over/under-confidence and $T^* = 1/\kappa$ follow immediately, and I prove a matching ECE bound. (C2) **Extensions** to the self-loop GCN operator, to K classes via a signal coefficient $c_K(h)$, and to covariate shift. (C3) **A structural corollary** (global-temperature optimality under homogeneous shift) that *predicts*, ahead of running any experiment, a negative result: per-node recalibration cannot help unless shift is heterogeneous across nodes. (C4) **A method, a reduction, and an honest negative result** (Sections 5–6): I turn the theory into STAC, a source-free, label-free recalibrator, and show the whole task reduces to estimating one scalar (target accuracy) from unlabeled data; a method built on standard estimators halves the gap on synthetic shifts but is unreliable across five real graphs, while a single oracle temperature recovers calibration on all of them. Label-free accuracy estimation under shift is the sole remaining bottleneck. (C5) **A positioning of the theory against the closest empirical alternatives** (Section 2, Table 1): I situate the closed-form account against recent empirical GNN-reliability methods [8, 9] and theory-grounded graph domain adaptation [10], clarifying which notion of reliability each provides and how they could be combined.

The remainder of this paper is organized as follows: Section 2 positions this work against the calibration, distribution-shift, and label-free-estimation literatures and summarizes eleven closely related methods along the axes that matter for deployment (Table 1). Section 3 fixes notation and the CSBM I analyze, and motivates why a linear GNN on a CSBM is the right level of abstraction. Section 4 states and proves the calibration-slope theorem and its extensions. Section 5 turns the theory into STAC, a source-free, label-free recalibration method. Section 6 validates every theoretical claim numerically and evaluates STAC on synthetic shifts and five real graphs. Section 7 discusses the implications, is honest about limitations, and lays out the open problems the theory exposes. Section 8 concludes. Appendix Appendix A gives full proofs of every theoretical result.

2. Related work

2.1. Calibration of neural networks

Modern networks trained with cross-entropy are systematically miscalibrated, typically over-confident on natural images [4]. Temperature scaling [4], dividing the logits by a single scalar fit by minimizing NLL on a labeled validation set, remains the most widely used post-hoc fix, extended to class-wise and Dirichlet variants [11]. Expected calibration error (ECE) and reliability diagrams [12] are the standard diagnostic; large-scale studies [13] show calibration is sensitive to architecture, augmentation, and training length even in-distribution, and training-time remedies such as focal loss [14] trade accuracy for calibration by reshaping the loss directly. Ovadia et al. [15] showed that all of these fixes degrade under distribution shift on image classifiers; the present paper supplies, for GNNs, the mechanism that was left as an empirical observation there.

2.2. In-distribution calibration of GNNs

GNNs miscalibrate in ways that are structurally distinct from image classifiers: rather than being uniformly over-confident, they are frequently under-confident in-distribution, and the size of the gap correlates with graph properties such as node degree and homophily [2, 3]. A first family of methods learns a per-node temperature from

an auxiliary model trained on labeled data: CaGCN [2] regresses temperatures with a second GCN; GATS [3] attends over structural covariates identified as drivers of miscalibration; GETS [5] ensembles several temperature-generating experts in a mixture-of-experts head; WATS [6] conditions the temperature on heat-kernel graph-wavelet features that encode multi-scale structure without relying on neighbours’ logits; SimCalib [7] ties calibration error directly to nodewise embedding similarity. A second family regularizes the training loss itself rather than post-processing logits, e.g. the graph calibration loss of Wang et al. [16]. All of these methods are trained and validated on data from the *same* distribution as the test graph; none models, or is evaluated under, distribution shift, and, as I discuss in Section 7.2, every one of them inherits the failure mode of ordinary temperature scaling once the target distribution moves. A related line quantifies predictive *uncertainty* rather than calibration per se, e.g. JuryGCN’s jackknife variance estimates [17]; such estimates are informative but do not by themselves specify a recalibration temperature.

2.3. GNN reliability under distribution shift

The gap this paper targets, calibration *under shift*, has so far been approached empirically rather than analytically. Trivedi et al. [8] document, across covariate, concept, and graph-size shifts, that increased GNN expressivity does not by itself improve confidence-indicator quality, and propose G- Δ UQ, a stochastic-anchoring scheme that estimates *epistemic* uncertainty from a single model and uses it to modulate confidence; this consistently improves out-of-distribution ECE relative to Monte Carlo dropout and vanilla temperature scaling, but offers no closed-form account of *why* shift moves calibration or of what the optimal correction is, and it requires retraining the base GNN with a stochastic-centering objective rather than recalibrating a frozen model. Huang et al. [9] take a complementary, distribution-free route: conformalized GNNs (CF-GNN) wrap any trained GNN with a topology-aware conformal predictor that provably achieves a target marginal coverage, trading the pointwise notion of calibration this paper studies for a set-valued guarantee that holds under exchangeability but is not designed to track *how* miscalibration grows as homophily or feature noise change. My closed-form slope κ is, to my knowledge, the first result that predicts the sign, magnitude, and optimal correction of GNN miscalibration under shift analytically, and the first to explain *why* a single global temperature should be expected to suffice.

2.4. Distribution shift and out-of-distribution generalization on graphs

A separate and much larger literature targets *accuracy*, rather than calibration, under graph distribution shift. Invariance-based training regularizes representations to be stable across environments (EERM [18], SR-GNN [19]), out-of-distribution generalization is benchmarked systematically by GOOD [20], and OOD-GNN [21] proposes an architecture-level generalization bound. You et al. [10] derive a model-based risk bound for graph domain adaptation and use it to design a spectral regularizer (GDA-SpecReg), theory-grounded in the same spirit as my approach, but aimed at bounding the target *error*, not at characterizing the resulting *confidence-accuracy* gap, and it requires access to the source graph at adaptation time. GTrans [22] adapts the graph itself (features and edges) at test time rather than the model, but again optimizes accuracy. None of these methods produce, or are evaluated on, calibrated probabilities.

2.5. Source-free and test-time adaptation

Outside calibration, source-free test-time adaptation removes the requirement of access to source data at deployment by adapting model parameters using only the target input distribution: entropy minimization (Tent [23]), self-supervised test-time training [24] (specialized to graphs by Wang et al. [25]), and source-hypothesis transfer (SHOT [26]) all optimize an unsupervised proxy for *accuracy* on the target data. STAC (Section 5) is source-free in the same sense (only a scalar threshold is carried from the source model), but optimizes a *calibration* objective rather than accuracy, and, following Corollary 1, deliberately restricts the adaptation to a single global scalar under homogeneous shift rather than adapting model weights.

2.6. Label-free performance estimation under shift

ATC [27] and disagreement-based generalization-gap estimators [28] predict a model’s accuracy on unlabeled, shifted data from confidence statistics and prediction disagreement under perturbation, respectively. Theorem 1 shows that these estimators supply exactly the missing ingredient a calibration method needs (a single target-accuracy scalar pins down $T^* = 1/\kappa$), and Section 6 shows this is simultaneously STAC’s strength (it reduces recalibration to one well-studied sub-problem) and its weakness (existing estimators are not accurate enough under graph shift; Section 7.4).

Table 1: **Positioning against closely related work.** “Test-distr. labels” asks whether the method needs labeled data assumed to represent the distribution it is applied to (all in-distribution calibrators do; STAC and the oracle baseline do not). “Shift-aware” asks whether the method’s design or analysis explicitly targets a *change* of distribution. “Granularity” is the unit of correction. “Theory” records the type of formal guarantee, if any.

Method	Year	Test-distr. labels	Shift-aware	Granularity	Theoretical guarantee
Temperature scaling [4]	2017	Yes	No	Global	Asymptotic NLL-optimality
CaGCN [2]	2021	Yes	No	Per-node	None
GATS [3]	2022	Yes	No	Per-node	None
CF-GNN [9]	2023	Yes	No	Sets (per-node)	Coverage (conformal)
GDA-SpecReg [10]	2023	No	Yes	Global (bound)	Risk bound (accuracy)
SimCalib [7]	2024	Yes	No	Per-node	None
G- Δ UQ [8]	2024	Yes (retrain)	Partial (empir.)	Per-node	None
GETS [5]	2025	Yes	No	Per-node	None
WATS [6]	2025	Yes	No	Per-node	None
STAC (this work)	n/a	No	Yes	Global	Closed form (κ, ECE bound)

2.7. Homophily-aware and physics-inspired GNN architectures

My signal coefficient $c_K(h)$ recovers the homophily threshold that motivates a family of spectral and physics-inspired GNN architectures designed to remain accurate across the homophily spectrum: FAGCN [29] learns a signed low/high-pass filter, GPR-GNN [30] learns signed propagation weights, and GraphCON [31] and A-DGN [32] use oscillatory/anti-symmetric ODE dynamics to propagate information without the over-smoothing that erodes signal at low homophily. These works change the *architecture* to remain accurate across h ; I hold the architecture fixed and instead show how h governs the confidence-accuracy gap of whatever architecture produced the logits, via $c_K(h)$ in Proposition 2. The two views are complementary: a homophily-robust architecture changes the operating point, but the calibration slope, and the optimality of a single temperature, follow from Theorem 1 regardless of which architecture produced the logits, consistent with my real-graph experiments, which calibrate SIGN/logistic-regression features, not the linear aggregation of Section 3 directly.

2.8. Summary and positioning

Table 1 summarizes the positioning of this paper against the eleven most closely related methods along the axes that determine deployability under shift: whether the method needs labels that are assumed representative of the test distribution, whether it explicitly models distribution shift, the granularity of the correction (a single global scalar vs. a per-node function), and whether it comes with a theoretical guarantee. Only the present work combines a closed-form, shift-aware theory with a fully label-free correction; the price, made explicit in Section 7.4, is that its practical reliability inherits the reliability of whichever label-free accuracy estimator is plugged in.

3. Problem setup

3.1. Problem statement

I am given a source graph $G_s = (A_s, X_s, Y_s)$ with node labels, on which a GNN f is trained and calibrated (e.g. by ordinary temperature scaling on a source validation split, so f is calibrated on G_s). At test time I observe a target graph $G_t = (A_t, X_t)$ from the *same* feature/label space but a *different* distribution (different homophily, different feature noise, or both), with *no* target labels and no further access to G_s or Y_s . The model f is frozen: re-training is not assumed to be an option in the deployment settings I target (streaming graphs, on-device inference, or simply the cost of retraining a production model). The question I answer is: how does the confidence-accuracy relationship of f ’s frozen predictions change from G_s to G_t , and how can it be corrected without labels?

3.2. Contextual stochastic block model

Nodes and labels. N nodes are partitioned into K balanced classes with labels y_i i.i.d. uniform on $\{1, \dots, K\}$. **Features.** $x_i = \mu_{y_i} + \xi_i$ with $\xi_i \sim \mathcal{N}(0, \sigma^2 I)$ i.i.d., class means $\|\mu_c\| = r$, $\sum_c \mu_c = 0$ (binary case: $\mu_{\pm} = \pm r\hat{\mu}$). The scale r relative to the noise level σ is the **signal-to-noise ratio** $\rho = r^2/\sigma^2$: large ρ means classes are easy to separate from features alone. **Edges and homophily.** Edges are intra-class with probability p and inter-class

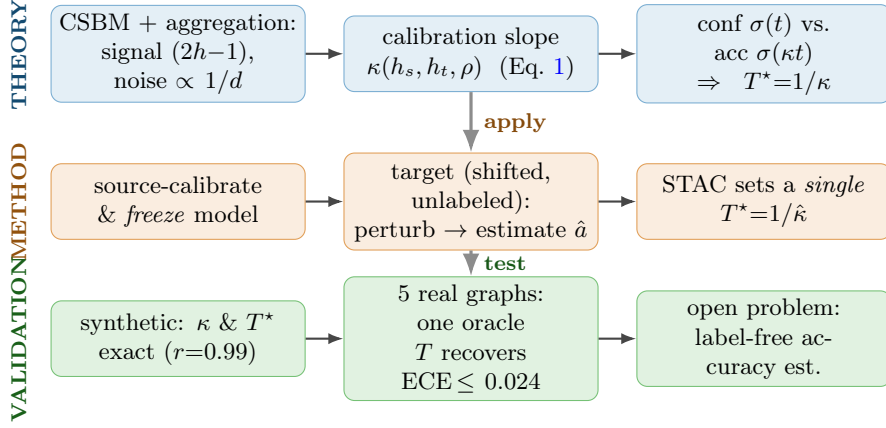


Figure 2: **Overview.** *Theory* (top): on a CSBM, aggregation scales the class signal by $(2h-1)$ and the noise by $1/d$, giving a closed-form calibration slope κ ; the model’s confidence is $\sigma(t)$ but its accuracy is $\sigma(\kappa t)$, so the optimal temperature is $T^*=1/\kappa$. *Method* (middle): freeze the source-calibrated model and, on the shifted unlabeled target, estimate accuracy from perturbations to set one temperature (STAC, Section 5). *Validation* (bottom): the closed form is exact in simulation and one oracle temperature recovers calibration on five real graphs; the only bottleneck is label-free accuracy estimation (Section 7.4).

with probability q ; the **edge homophily** $h = p/(p + q(K-1))$ is the probability that a uniformly random edge joins two same-class nodes: $h > 1/K$ is homophilic, $h < 1/K$ heterophilic, and $h = 1/K$ is the point at which class structure is invisible to a same-class-counting statistic. This is the standard contextual SBM used to study information-plus-structure learning problems [33].

3.3. Linear GNN and calibration

I analyze mean aggregation $z_i = \frac{1}{d_i} \sum_{j \sim i} x_j$ and the symmetric-normalized GCN operator $\hat{A} = \tilde{D}^{-1/2}(A+I)\tilde{D}^{-1/2}$ [34], followed by a linear classifier calibrated on the source graph; multi-hop SIGN [35, 36] features are used on real data in Section 6. With logit δ_i , confidence is $\sigma(|\delta_i|)$ and the prediction is correct iff $\text{sign}(\delta_i) = y_i$; ECE bins predictions by confidence and averages $|\text{acc} - \text{conf}|$ [4, 12]. Temperature scaling maps $\delta \mapsto \delta/T$ and never changes predictions, so it cannot help or hurt accuracy, only calibration.

3.4. Why a linear GNN on a CSBM?

A skeptical reader may ask why I restrict the closed-form analysis to a linear classifier on Gaussian CSBM features rather than a deep, nonlinear GNN. I follow a long tradition in learning theory (of which the CSBM itself is a canonical graph instance [33]) of trading architectural realism for a model simple enough to solve exactly, then checking numerically how far its predictions travel toward the regime of interest. Three considerations support this choice here. First, mean/GCN aggregation followed by a linear read-out is not merely a toy: it is the exact computation performed by SGC [36] and, feature-wise, by SIGN [35], both competitive with deeper GNNs on the heterophilous benchmarks I use in Section 6, so the model class I analyze is a real, widely used GNN family, not merely an approximation of one. Second, the object I characterize, the map from logit magnitude to accuracy, depends on the *aggregation step*, which mixes same- and different-class neighbours in a homophily-dependent ratio, far more than on the specific nonlinearity of the classifier head; this is why the closed-form slope, derived for a linear head, tracks a self-loop GCN’s simulated slope closely (Section 6.1, Fig. 6) and a logistic-regression head on SIGN features on five real graphs (Section 6.4). Third, and most importantly, every qualitative claim the theory makes (the sign of over/under-confidence, the existence of a single optimal temperature, the non-benefit of per-node calibration under homogeneous shift) is falsifiable and is checked against simulation and real data throughout Section 6; I treat the tractable model as a source of falsifiable, quantitative predictions about a real computation (mean/GCN aggregation plus a linear or near-linear head), not as a claim about mechanism inside arbitrarily deep, nonlinear GNNs.

4. Theoretical analysis

All proofs are in Appendix A; results hold in a large-degree Gaussian regime and are confirmed numerically (Section 6). Figure 2 previews how the three parts of the paper (theory, method, and validation) fit together.

4.1. The aggregated discriminant coordinate

Lemma 1 (Aggregated coordinate). *Under mean aggregation, conditional on $y_i = +1$ and degree d_i , the discriminant coordinate $g_i = \hat{\mu}^\top z_i$ has mean $(2h-1)r$ and variance $(4h(1-h)r^2 + \sigma^2)/d_i$, and is asymptotically Gaussian as $d_i \rightarrow \infty$.*

Remark 1 (Reading the lemma). The mean $(2h-1)r$ is exactly the excess of same-class over different-class neighbours, scaled by the class-mean separation r ; it is positive for homophilic graphs ($h > 1/2$), zero at $h = 1/2$ where a neighbour is a coin flip over the two classes, and negative for heterophilic graphs, where aggregation systematically pulls a node’s representation *toward the wrong class*. The variance combines two independent noise sources, label randomness among neighbours ($4h(1-h)r^2$) and feature noise (σ^2), and both shrink as $1/d_i$: high-degree nodes average out noise exactly as one would expect from a sample mean of d_i roughly independent terms. Gaussianity is not an approximation tacked on for convenience; it is a genuine Lindeberg central-limit effect over the d_i neighbours, which is why the closed forms below are accurate already at moderate degree ($d \approx 20$, Section 6).

4.2. The calibration slope

Theorem 1 (Calibration slope). *Let the logit be $\delta = a g$, so $\delta \mid y=+1 \sim \mathcal{N}(m, s^2)$ with $m = a(2h-1)r$, $s^2 = a^2(4h(1-h)r^2 + \sigma^2)/d$, and $\delta \mid y=-1 \sim \mathcal{N}(-m, s^2)$. With $\kappa := 2m/s^2$, for every t*

$$\Pr(\text{correct} \mid \delta=t) = \sigma(\kappa t), \quad \text{confidence}(t) = \sigma(t).$$

Hence the model is over-confident iff $\kappa < 1$, under-confident iff $\kappa > 1$, calibrated iff $\kappa = 1$; the NLL-optimal temperature is $T^ = 1/\kappa$; and with a fixed by source calibration ($\kappa(h_s)=1$),*

$$\kappa(h_t) = \frac{(2h_t-1)(1 + 4h_s(1-h_s)\rho)}{(2h_s-1)(1 + 4h_t(1-h_t)\rho)}, \quad (1)$$

independent of degree.

Remark 2 (Reading Eq. (1)). Equation (1) says homophily controls the *sign and strength* of the aggregation signal, $(2h-1)$: moving h_t toward $\frac{1}{2}$ shrinks the signal faster than the variance, so $\kappa < 1$ and the frozen model becomes over-confident; a target more homophilous than the source ($h_t > h_s$, both $> 1/2$) gives $\kappa > 1$ and the model becomes under-confident, because the same logit magnitude that was appropriate for a noisier source aggregation now understates how reliable the cleaner target signal actually is. The formula is antisymmetric in a useful sense: $\kappa(h_s) = 1$ by construction, and κ crosses zero exactly when $h_t = 1/2$, where the aggregated signal vanishes and no logit magnitude, however large, can be predictive, an early warning that the bound in Proposition 4 becomes uninformative there. The dependence on ρ enters only through the combination $4h(1-h)\rho$, i.e. SNR matters only insofar as it competes with the same-/different-class mixing variance that homophily itself controls; at $\rho \rightarrow \infty$ (noiseless features) Eq. (1) reduces to the purely structural ratio $(2h_t-1)/(2h_s-1)$.

Corollary 1 (Global temperature is optimal). *Under homogeneous shift (a single κ), $T^* = 1/\kappa$ makes the model equal to the true conditional, attaining minimum population NLL and zero population ECE. Any per-node temperatures $\{T_i\}$ with $T_i \neq T^*$ on a positive-measure set strictly increase NLL. Per-node recalibration can therefore only help under heterogeneous shift.*

Remark 3 (Why this corollary matters operationally). Corollary 1 is a negative result stated as a positive one: it says that *any* extra modeling effort spent on a per-node temperature function is provably wasted under homogeneous shift, no matter how expressive the per-node model is, because the population-optimal per-node solution is already the constant function T^* . This is a strong, falsifiable prediction about methods like CaGCN and GATS (Section 2), which fit a per-node temperature *even when* the shift they are correcting for is homogeneous; Section 6.3 tests the prediction directly with a dedicated ablation (STAC-global vs. STAC-full) and finds it holds: the per-node stage changes error-detection ranking only marginally, and not more than an ablation that discards graph structure entirely.

4.3. Extensions

Proposition 1 (Self-loop GCN). *For $\hat{A} = \tilde{D}^{-1/2}(A+I)\tilde{D}^{-1/2}$, with $A(h) = 1 + d(2h-1)$ and $B(h) = 4dh(1-h)\rho + (d+1)$, $\kappa_{SL}(h_t) = A(h_t)B(h_s)/[A(h_s)B(h_t)]$, which reduces to (1) as $d \rightarrow \infty$. The self term makes the signal survive down to $h = (1-1/d)/2 < \frac{1}{2}$, formalizing why self-loops help under heterophily.*

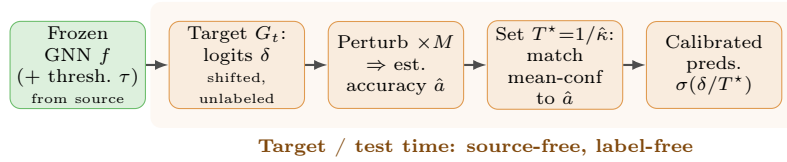


Figure 3: **STAC**. At test time the frozen source model is applied to the shifted, unlabeled target graph; a few perturbed passes yield a label-free accuracy estimate $\hat{a}$, which fixes a single temperature $T^* = 1/\hat{a}$ so that mean confidence matches $\hat{a}$. No target labels and no source data are used.

Remark 4. The self-loop term adds a deterministic same-class contribution ($+r$ with certainty, since a node is always its own class) that does not exist for a node’s noisy neighbours. This shifts the zero-crossing of the aggregated signal from exactly $h = 1/2$ (mean aggregation) to $h = (1-1/d)/2$, slightly below $1/2$: self-loops buy a small but degree-dependent margin of heterophily tolerance before the signal vanishes, consistent with the common empirical observation that adding self-loops helps GCNs on mildly heterophilic graphs.

Proposition 2 (K classes). *With symmetric simplex means, conditional on $y_i = c$, $\mathbb{E}[z_i] = c_K(h) \mu_c$ with $c_K(h) = (hK-1)/(K-1)$ (zero at $h = 1/K$, equal to $2h-1$ for $K=2$). Homophily shift rescales the whole logit vector by $c_K(h_t)/c_K(h_s)$, so a single temperature suffices.*

Remark 5. $c_K(h)$ is the natural K -class generalization of $(2h-1)$: a same-class neighbour (probability h) contributes $+\mu_c$, and a different-class neighbour (probability $1-h$, uniform over the remaining $K-1$ classes) contributes $-\mu_c/(K-1)$ in expectation because the simplex means sum to zero. The zero-crossing moves from $1/2$ to $1/K$ as the number of classes grows: the more classes there are, the less homophily is needed before a random neighbour is more likely to be a different class than not (Fig. 4 in Section 6 illustrates this schematically). Because the rescaling by $c_K(h_t)/c_K(h_s)$ is common to every class direction, it acts on the full K -dimensional logit vector exactly as a scalar temperature would, which is why a single global temperature remains sufficient even in the multi-class case, as confirmed numerically for $K = 3, 4, 5$ in Section 6.1.

Proposition 3 (Covariate shift). *At fixed h , if the target feature-noise variance changes $\sigma^2 \mapsto \sigma^2 + \tau^2$ ($\gamma = \tau^2/\sigma^2$), then with source calibration $\kappa_{\text{cov}} = \frac{4h(1-h)\rho+1}{4h(1-h)\rho+1+\gamma} < 1$ and $T^* = 1 + \gamma/(4h(1-h)\rho + 1) > 1$: covariate shift always induces over-confidence, monotonically in γ .*

Remark 6. Unlike homophily shift, covariate shift is one-directional: because extra feature noise can only inflate the logit variance s^2 (it cannot change the signal m , which depends only on the class means and homophily), κ_{cov} can only fall below 1. This is a clean, falsifiable prediction (covariate shift never causes under-confidence), which Section 6.5 confirms on five real graphs: the confidence-accuracy gap grows monotonically, and only in the over-confident direction, as the relative feature-noise strength increases.

Proposition 4 (ECE bound). *$\text{ECE} \leq \frac{1}{4}|\kappa - 1| \mathbb{E}|\delta|$; it vanishes as $\kappa \rightarrow 1$ and grows with the logit magnitude.*

Remark 7. The bound follows from the $\frac{1}{4}$ -Lipschitz constant of the sigmoid and turns Theorem 1’s *pointwise* statement into a single scalar diagnostic: because $\mathbb{E}|\delta|$ is observable from the model’s own logits on the target graph *without labels*, this bound gives a fully label-free, if conservative, upper estimate of how miscalibrated a frozen model can possibly be under a shift of a given severity $|\kappa - 1|$, useful as an early-warning signal even before attempting the recalibration of Section 5. The bound is tightest, and the recalibration correspondingly most consequential, exactly when confident predictions ($|\delta|$ large) coincide with a shift severe enough to move κ far from 1; Section 6 confirms empirically that the bound is tight near the source and looser, but never violated, far from it.

5. STAC: a source-free, label-free recalibration method

Theorem 1 says the entire job is to estimate one number, $T^* = 1/\kappa$, from *unlabeled* target data (Fig. 3, Algorithm 1). I call the resulting method **STAC** (Structure-aware Test-time Adaptive Calibration). STAC (i) estimates target accuracy $\hat{a}$ from unlabeled data by combining thresholded confidence [27] with prediction disagreement under input/edge perturbations [28], then (ii) chooses the single T so that mean confidence equals $\hat{a}$. The method is source-free (only a scalar threshold is carried from training), adds $O(1)$ parameters, and needs no GPU.

Algorithm 1 STAC: label-free test-time recalibration (global stage)

Require: frozen model f , source threshold τ , target graph G_t , passes M

- 1: $\delta \leftarrow$ logits of f on G_t ; $\hat{a}_{\text{ATC}} \leftarrow \text{frac. of } \max \sigma(\delta) \geq \tau$
 - 2: run M perturbed passes; $\hat{a}_{\text{GDE}} \leftarrow 1 - \text{mean pairwise disagreement}$
 - 3: $\hat{a} \leftarrow \min(\hat{a}_{\text{ATC}}, \hat{a}_{\text{GDE}})$
 - 4: $T^* \leftarrow \arg \min_T (\text{mean}_i \max_k \sigma(\delta_i/T)_k - \hat{a})^2$
 - 5: **return** calibrated probabilities $\sigma(\delta/T^*)$
-

Complexity. Algorithm 1 requires M additional forward passes of the (frozen) model over the target graph (no gradients, no retraining), plus an $O(M^2)$ pairwise-disagreement computation per node and a one-dimensional line search for T . In my experiments $M = 6$ – 10 suffices, and the whole procedure runs in seconds on CPU for graphs with tens of thousands of nodes; the dominant cost is the M forward passes, which is the same order of cost as computing M -sample Monte Carlo dropout uncertainty, but here applied at test time to a deterministic frozen model via input and edge perturbations rather than at training time to a stochastic one.

Optional per-node stage. A second, optional stage modulates the global temperature per node, $T_i = T_{\text{glob}} \exp(c \tanh(\phi_i^\top w + b))$, where ϕ_i collects structure-aware signals (feature- and edge-perturbation instability, homophily-corrected neighbour-agreement, prediction entropy, margin, and log-degree), fit by minimizing the same label-free binned calibration loss with a small ridge penalty on w . By Corollary 1, this stage is *not expected to help* under homogeneous shift, and I include it purely as an empirical test of that prediction: an ablation without structural signals (logits only) isolates whether any residual benefit comes from graph structure specifically or merely from having a higher-capacity per-node function. Section 6.3 reports the outcome of this ablation.

Relation to the theory. The reduction to a single scalar $\hat{a}$ is not a simplifying choice I make for convenience; it is what Theorem 1 and Corollary 1 *prove* is sufficient under homogeneous shift. This is what distinguishes STAC’s global stage from the per-node calibrators surveyed in Section 2: those methods use a per-node function because they have no theoretical reason to expect a global scalar to be optimal, whereas STAC uses a global scalar *because* the theory says it is optimal, and only reaches for a per-node correction as a deliberately falsifiable ablation.

6. Experiments

All experiments run on CPU using only NumPy/SciPy/scikit-learn; code and data are released. I report ECE (15 bins) throughout.

6.1. Theory validation (synthetic)

On a binary CSBM ($h_s=0.8, \rho=1$), the measured slope $\kappa = 1/T_{\text{oracle}}$ matches the closed form (1) across $h_t \in [0.55, 0.9]$, capturing both over- and under-confidence, with $\kappa \approx 1$ at the source (Fig. 5, left); ECE grows from 0.004 to 0.25 as h_t leaves h_s and T^* returns it to ≤ 0.02 (right). Figure 4 illustrates schematically why homophily has this effect: aggregation averages same- and different-class neighbours, and the resulting signal along the true class direction is scaled by $c_K(h)$, which vanishes and flips sign at $h = 1/K$ (Proposition 2). The self-loop slope κ_{SL} matches the GCN simulation and is closer to it than the mean-aggregation formula in the heterophilous regime (Fig. 6, left); the signal coefficient $c_K(h)$ matches the measured value *exactly* for $K = 3, 4, 5$ (right). The bound of Proposition 4 holds throughout (Fig. 7) and, as anticipated in the remark following it, is tightest near the source homophily and looser, but never violated, far from it.

6.2. The closed form predicts the optimal temperature

Beyond signs, the theory predicts the *value* $T^* = 1/\kappa$. Over a grid of source/target homophilies and SNRs on the binary CSBM, the predicted T^* matches the oracle temperature measured on labeled target data with Pearson $r = 0.99$ and a mean absolute error of 0.11 temperature units (Fig. 8). The closed form is thus quantitatively predictive, not merely directional.

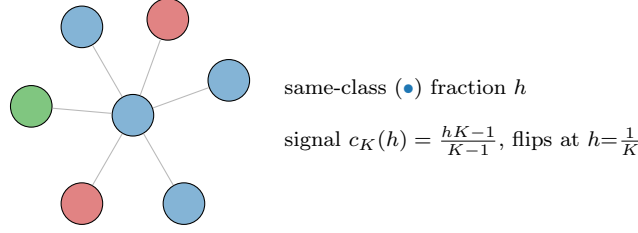


Figure 4: **Why homophily controls calibration.** Aggregation averages same- and different-class neighbours; the resulting signal along the true class is scaled by $c_K(h)$, which vanishes and flips sign at $h = 1/K$.

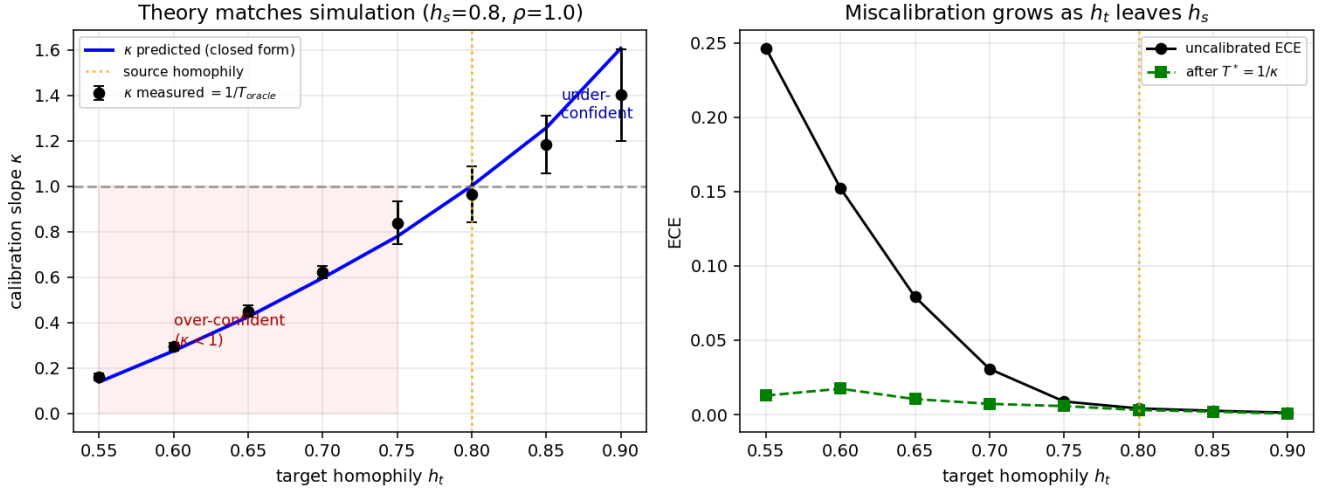


Figure 5: **Theorem 1.** Closed-form slope vs. measured $1/T_{\text{oracle}}$ (left); ECE before/after T^* (right).

6.3. STAC under shift (synthetic)

On CSBM node classification with a trained 2-layer GCN, across feature, homophily, and heterogeneous shifts, STAC’s global stage roughly *halves* the ECE gap versus source-temperature scaling and approaches the oracle. Figure 9 makes the mechanism concrete under increasing covariate-shift intensity. The left panel shows ECE for four methods: the uncalibrated model degrades steadily as shift grows; source-TS, fit once on the source validation split, barely helps and at high shift intensity is close to indistinguishable from no calibration at all, because it targets the *source* accuracy rather than the falling target accuracy; STAC-global tracks roughly half-way between source-TS and the oracle at every shift level; the oracle (which uses target labels solely to fit T , never to retrain) stays flat and low throughout. The right panel diagnoses *why*: it plots the true target accuracy against the two candidate proxies a calibrator could use. The ‘Source-TS assumption’ (dotted) is exactly the source validation accuracy, which is shift-invariant by construction and therefore departs from the true target accuracy (solid) as shift grows; this gap is the empirical face of κ moving away from 1 as shift intensity increases, and it is why source-fitted calibrators fail under shift (Section 7.2). STAC’s label-free accuracy estimate (dashed) tracks the true target accuracy far more closely than the source assumption does, but not perfectly; the residual gap between the dashed and solid curves is exactly what limits STAC-global’s ECE relative to the oracle in the left panel, and it is the same gap that becomes unmanageable on real, heterophilous graphs in Section 6.4.

Under the heterogeneous (‘mixed’) shift, where only a random half of target nodes receive a strong feature perturbation, so no single κ describes the whole graph, a single temperature (STAC-global, identical to source-TS by construction once its level is fixed) cannot, by definition, change the *ranking* of nodes by confidence, so its error-detection AUROC exactly matches the uncalibrated model’s. Only the optional per-node stage (STAC-full, Section 5), which modulates T_i using structure-aware signals, can move this ranking; in my runs it does so only marginally, and an ablation without structural signals (logits only) performs comparably, indicating that, at least for

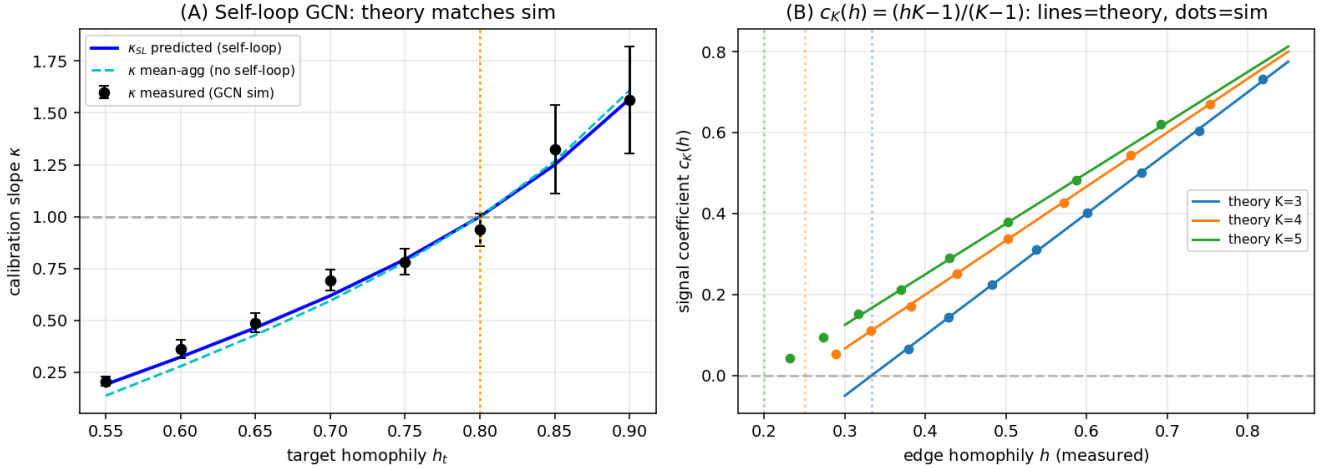


Figure 6: **Extensions.** Self-loop GCN slope κ_{SL} (left) and the K -class signal coefficient $c_K(h)$ (right): lines are theory, markers are simulation.

Table 2: ECE ($\downarrow$) on real graphs under covariate shift (3 splits).

graph	h	Uncal.	Src- T	STAC	Oracle
roman-empire	0.05	0.145	0.206	0.200	0.022
amazon-ratings	0.38	0.085	0.075	0.432	0.024
tolokers	0.59	0.132	0.052	0.168	0.015
minesweeper	0.68	0.123	0.072	0.128	0.020
questions	0.84	0.024	0.006	0.028	0.007

the perturbation signals I tried, graph structure does not add error-detection information beyond what the model’s own logits already carry. This is precisely the falsifiable prediction of Corollary 1 put to the test: a single global scale is a genuinely different problem from finding a useful per-node ranking signal, and my attempt at the latter with off-the-shelf structural features does not yet succeed; I return to this as an open problem in Section 7.6. For reference, under feature shift (target accuracy 0.61): Uncalibrated ECE 0.203, Source-TS 0.191, STAC-global 0.100, Oracle 0.024.

6.4. Real graphs across the homophily spectrum

I evaluate on five real graphs [37], namely roman-empire ($h=0.05, K=18$), amazon-ratings (0.38, 5), tolokers (0.59, 2), minesweeper (0.68, 2) and questions (0.84, 2), under a fixed relative covariate shift (Fig. 10, Table 2). The robust finding matches the theory: a single *oracle* temperature recovers calibration on *every* graph (ECE 0.015–0.024), across the full homophily range and multi-class settings. However the *label-free* STAC is not reliable here: because the accuracy estimate overshoots under shift (e.g. amazon-ratings $\hat{a}=0.81$ vs. true 0.38), it can worsen ECE (0.085 $\rightarrow$ 0.432). In-distribution calibrators [2, 3, 5–7] are source-fitted and thus inherit the failure of source-TS under shift; empirical uncertainty-quantification methods designed with shift in mind, such as G- Δ UQ [8], could plausibly supply a better $\hat{a}$ here, since neither it nor STAC’s simple ATC/disagreement estimator was designed for the heterophilous, high- K regime of roman-empire; I see this as a promising, direct avenue for future work (Section 7.4) rather than a competing claim. I report STAC’s negative result plainly.

6.5. Covariate-shift strength on real graphs

Proposition 3 predicts that covariate shift makes the frozen model progressively over-confident, and *only* over-confident. Sweeping the relative feature-noise strength γ on all five graphs, the confidence–accuracy gap grows monotonically with γ and never changes sign (Fig. 11a; e.g. roman-empire $-0.07 \rightarrow +0.57$, minesweeper $+0.07 \rightarrow +0.29$), while a single oracle temperature keeps ECE low throughout (Fig. 11b). The nearly flat curve for questions ($h=0.84$, accuracy 0.97) is expected: Proposition 3 gives a gap $\propto \gamma/(4h(1-h)\rho+1)$, small when accuracy is high.

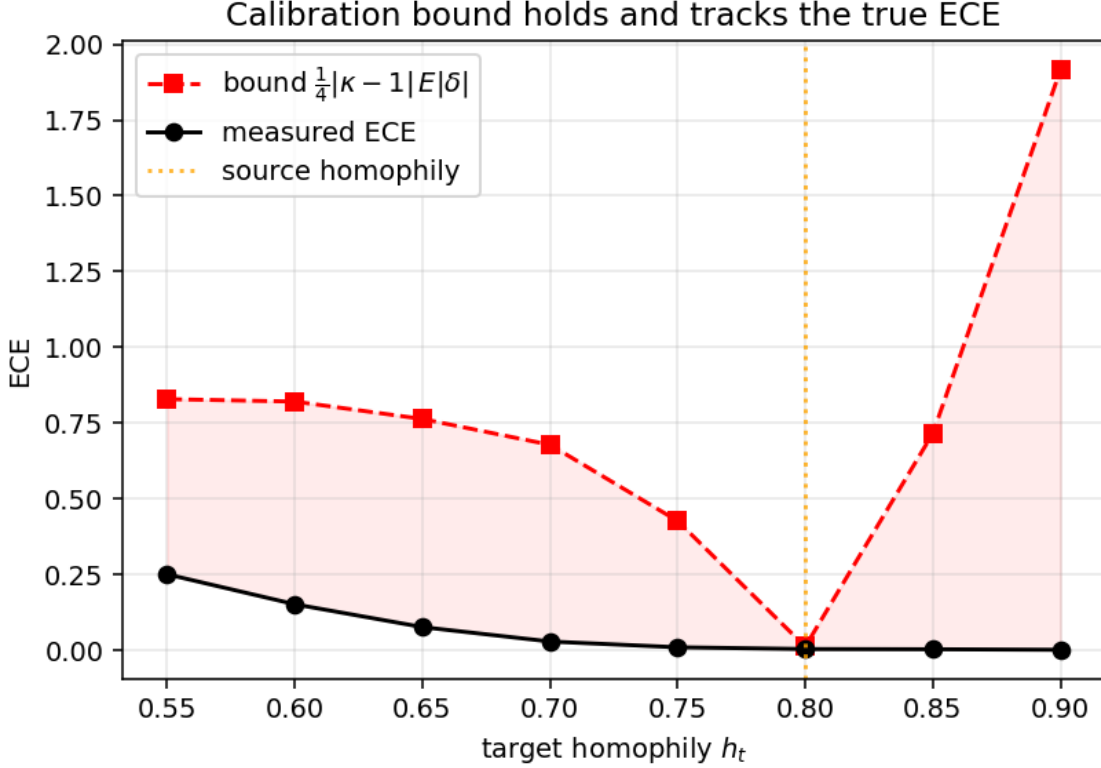


Figure 7: **Proposition 4.** The bound $\frac{1}{4}|\kappa - 1|E|\delta|$ upper-bounds the measured ECE and is tight near the source.

6.6. A different shift type: structural rewiring

To test breadth beyond covariate noise I rewire a fraction of the target edges. This miscalibrates several graphs (e.g. minesweeper ECE $0.07 \rightarrow 0.14$ at 75% rewiring), and a single oracle temperature again recovers calibration across all five (ECE ≤ 0.03 ; Fig. 12). For class-imbalanced graphs whose random-edge homophily is already high (questions, tolokors) rewiring changes homophily little and calibration is largely unaffected, consistent with the theory, in which the effect is mediated by the change in edge homophily.

7. Discussion, limitations, and open problems

I merge discussion and limitations deliberately: every limitation below is the direct, falsifiable counterpart of a claim made in Section 4, and reading them together is what makes the theory’s scope honest rather than merely optimistic.

7.1. The calibration slope as a diagnostic

Beyond motivating STAC, κ is useful on its own: Eq. (1) can be evaluated from an estimate of (h_s, h_t, ρ) alone, quantities that are far easier to estimate without labels than accuracy itself (homophily can be estimated from a label-propagation proxy or from the model’s own agreement statistics; SNR from feature clustering), to *predict* whether a deployed GNN is likely to be over- or under-confident on a new graph before any recalibration is attempted at all, using Proposition 4 as a conservative severity bound.

7.2. Why source-fitted calibrators fail under shift

Every in-distribution calibrator surveyed in Section 2 (temperature scaling itself, CaGCN, GATS, GETS, WATS, SimCalib) fits its correction (a scalar or a per-node function) by minimizing NLL on a labeled split assumed to represent the distribution the model will see at test time. Theorem 1 makes explicit *why* this assumption breaking is fatal: the fitted correction is optimal for $\kappa(h_s, h_s, \rho) = 1$ by construction, but the deployment-time slope is $\kappa(h_s, h_t, \rho) \neq 1$ whenever $h_t \neq h_s$, and no amount of additional labeled *source* data changes this, because the source split never observes h_t . This is precisely the mechanism Fig. 9 makes visible: source-TS’s assumption curve is flat

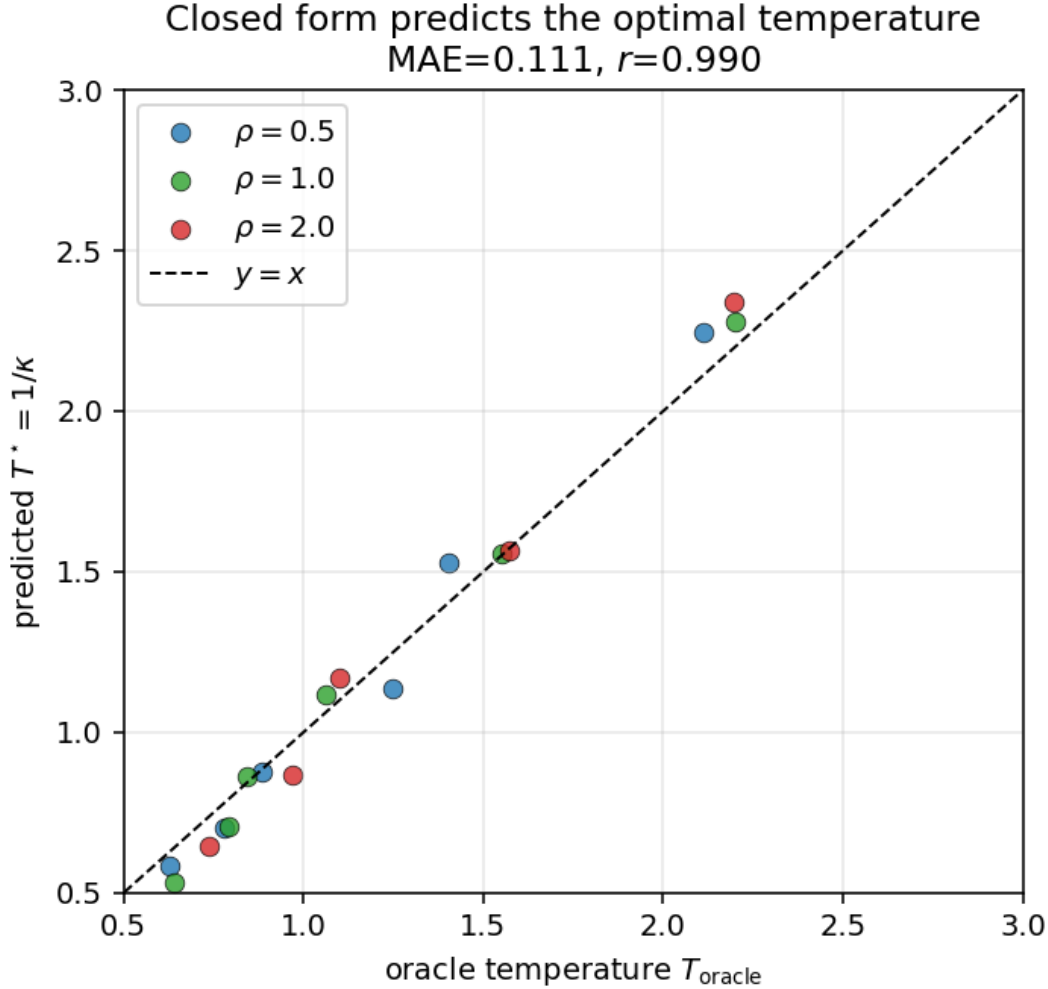


Figure 8: **Optimal temperature.** Predicted $T^* = 1/\kappa$ vs. the oracle temperature across (h_s, h_t, ρ) configurations; points hug $y=x$ ($r=0.99$, MAE 0.11).

by construction, while the true target accuracy moves. Methods with a per-node correction do not escape this: a per-node function fit on source data still targets $\kappa(h_s, h_s, \rho) = 1$ pointwise, uniformly across nodes, so it fails in exactly the same direction as global temperature scaling, merely with additional, source-fitted degrees of freedom that provide no protection against a shift in h .

7.3. When does per-node calibration help?

Corollary 1 localizes the value of per-node calibration precisely: it helps if and only if different nodes experience *different* effective slopes κ_i (heterogeneous shift), never as a matter of raw modeling capacity. This reframes the comparison in Section 2 between global and per-node calibrators: the question is not “does a more expressive per-node model calibrate better” but “is the shift itself homogeneous or heterogeneous”, which is a property of the deployment scenario, not of the calibrator. My heterogeneous-shift ablation (Section 6.3) is a first attempt to answer this operationally, and it suggests that answering it well requires per-node *shift-detection* signals that are more informative than the structural features I tried; see Section 7.6.

7.4. The crux is label-free accuracy estimation

This is the paper’s central limitation, and I state it as plainly as possible: STAC’s entire practical reliability rests on $\hat{a}$, and Section 6.4 shows that on real, heterophilous, high- K graphs (roman-empire, amazon-ratings), both ATC and disagreement-based estimators can be badly wrong, overshooting true accuracy by a wide margin, and thereby set T^* in the wrong direction. Two implications follow, one constructive and one methodological. Constructively,

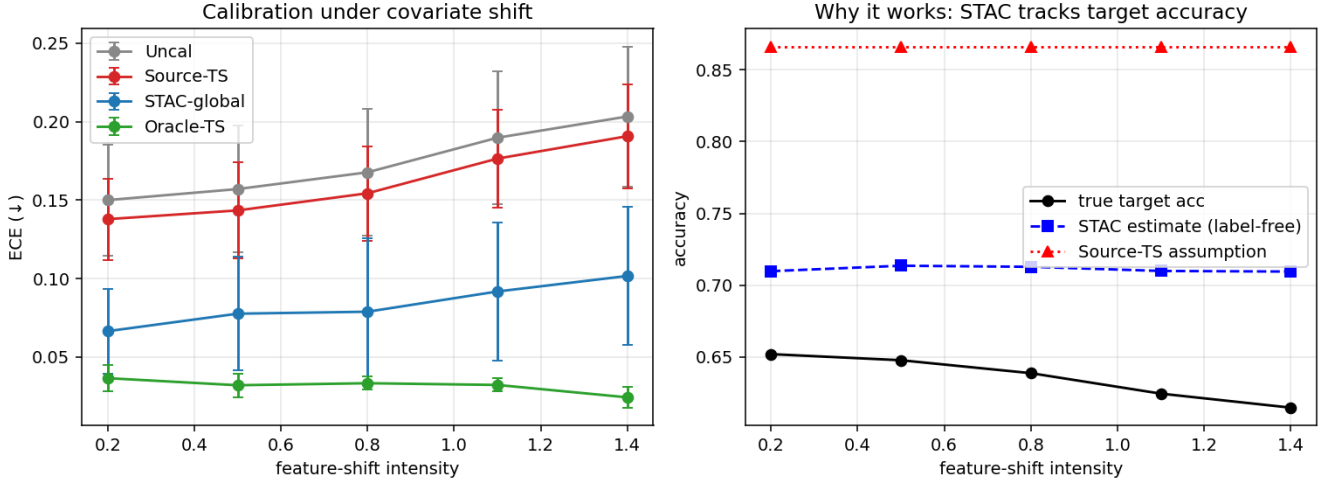


Figure 9: **Mechanism (synthetic feature shift).** *Left:* ECE vs. shift intensity for the uncalibrated model, source-TS, STAC-global, and the oracle; STAC-global consistently halves the source-TS gap. *Right:* the true target accuracy (solid) falls as shift grows, while the source-TS assumption (dotted, frozen at the source validation accuracy) does not move at all: this divergence is the mechanism behind source-TS’s failure; STAC’s label-free estimate (dashed) tracks the true accuracy much more closely, though not exactly.

the oracle results (Table 2) show the theory’s ceiling is high: *if* $\hat{a}$ were accurate, a single temperature would suffice everywhere I tested, from $h=0.05$ to $h=0.84$ and up to $K=18$ classes, so the return on improving label-free accuracy estimation specifically *for graphs* is large. Methodologically, ATC and disagreement were designed and validated primarily on covariate shift for i.i.d. image/text data [27, 28]; graphs add at least two confounds absent there: aggregation compounds shift across hops, so a node’s effective shift depends on its neighbourhood’s shift, not only its own features, and homophily itself is a further axis of shift that has no counterpart in the i.i.d. setting these estimators were built for. A concrete, falsifiable next step is a graph-native accuracy estimator that conditions disagreement on the local homophily estimated at the same node, or that combines STAC’s estimator with the epistemic-uncertainty signal of G- Δ UQ [8] (Section 2), which was explicitly validated under graph covariate and concept shift and could plausibly correct exactly the overshoot I observe on amazon-ratings and roman-empire.

7.5. Scope of the theory

The closed forms assume Gaussian class-conditional features and are asymptotic in degree d (a Berry–Esseen-type correction is $O(d^{-1/2})$); Section 6 shows this is already accurate at moderate degree ($d \approx 20$) but I do not claim a non-asymptotic, finite-degree guarantee. The CSBM is also a *fixed-homophily* model: real graphs mix a global homophily with substantial node-to-node heterogeneity, which is exactly the gap between the homogeneous-shift theory of Section 4 and the heterogeneous-shift ablation of Section 6.3. Finally, near chance accuracy (as on parts of roman-empire and amazon-ratings) the logistic approximation underlying Theorem 1 is least informative, precisely the regime where label-free accuracy estimation is also hardest (Section 7.4); these two limitations compound rather than being independent, which is why real-graph performance in Table 2 is weakest exactly where the pointwise theory is weakest.

7.6. Heterogeneous shift is the open per-node problem

Corollary 1 says *when* per-node calibration is needed (heterogeneous shift) but not *how* to detect, node by node, which effective κ_i applies; my attempt (Section 6.3) with generic structural perturbation signals moved error-detection AUROC only marginally above a structure-free baseline. I view this negative result as informative rather than discouraging: it narrows the open problem to finding per-node signals that specifically track *local* shift severity; candidates include local homophily estimated from a node’s immediate neighbourhood (directly motivated by Eq. (1)), a per-node analogue of the epistemic-uncertainty estimator of Trivedi et al. [8], or the per-node conformal width produced by Huang et al. [9], which is a different notion of per-node reliability (set size) that could plausibly correlate with κ_i without requiring labels.

7.7. Broader outlook

I see three natural combinations with the empirical literature surveyed in Section 2. First, G- Δ UQ’s per-model epistemic-uncertainty estimate [8] could serve directly as the per-node shift-severity signal that Section 7.6 shows is

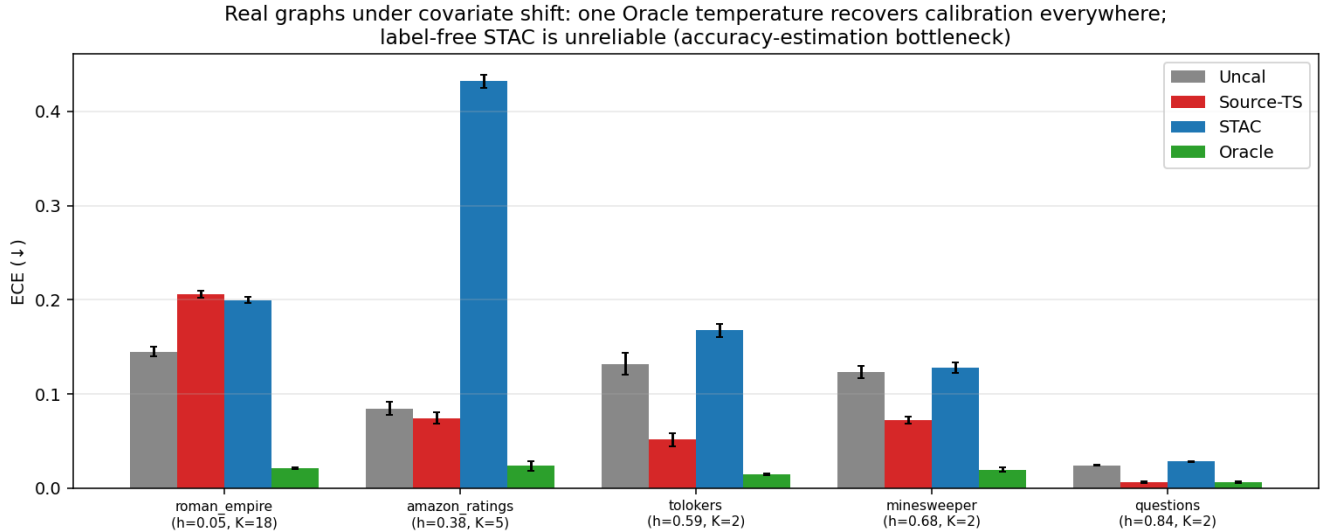


Figure 10: **Five real graphs under covariate shift.** One oracle temperature recovers calibration across the homophily spectrum; STAC is unreliable (accuracy-estimation bottleneck).

currently missing. Second, the distribution-free coverage guarantees of conformalized GNNs [9] address a set-valued, not pointwise, notion of reliability; combining a globally-corrected temperature with a conformal wrapper is plausible and would let the two guarantees (closed-form pointwise calibration and distribution-free coverage) compose rather than compete. Third, GDA-SpecReg’s risk bound [10] bounds target *error*; an accuracy bound of that kind, if tight enough, is itself a candidate label-free $\hat{a}$ for STAC, directly addressing Section 7.4.

8. Conclusion

This paper gives, to my knowledge, the first closed-form account of how distribution shift moves the confidence-accuracy relationship of a graph neural network: a single calibration slope κ , explicit in source/target homophily and feature signal-to-noise ratio, determines simultaneously the *direction* of miscalibration ($\kappa < 1$ over-confident, $\kappa > 1$ under-confident), its *magnitude* (via the proven ECE bound), and its *exact correction* ($T^* = 1/\kappa$, provably optimal at the population level and provably sufficient as a single global scalar under homogeneous shift). The theory is not a qualitative story: it predicts the measured slope and the oracle temperature quantitatively in simulation ($r = 0.99$), and its central operational consequence (a single temperature recovers calibration) holds on five real graphs spanning two orders of magnitude in homophily and up to 18 classes.

Turning the theory into a fully automatic, label-free recalibrator reduces to estimating one number, target accuracy, from unlabeled data; I showed this reduction is correct in that a good estimate (the oracle) suffices, but that today’s off-the-shelf estimators (ATC, disagreement) are not yet accurate enough on real, heterophilous graphs to realize the theory’s full promise. I see this not as a weakness of the theory but as its most useful output: it isolates exactly the sub-problem (shift-robust, graph-native accuracy estimation) on which future work should concentrate to convert closed-form understanding into a deployable, zero-label GNN calibrator, and Section 7 lays out concrete, falsifiable directions toward that goal, including combinations with recent empirical uncertainty-quantification and conformal-prediction methods for GNNs. More broadly, I hope the calibration-slope viewpoint offers a template for other structure-dependent reliability questions on graphs: write down the smallest model that reproduces a failure mode in closed form, derive its falsifiable consequences, and test how far its predictions travel toward real, deployed graph learning systems.

Appendix A. Assumptions and proofs

This appendix states the formal assumptions underlying Sections 4, and gives complete proofs of every theoretical claim in the main text. All results concern a linear GNN on a CSBM in a large-degree regime; finite-degree accuracy is discussed in Remark 8 and confirmed numerically in Section 6 (Figs. 5–6).

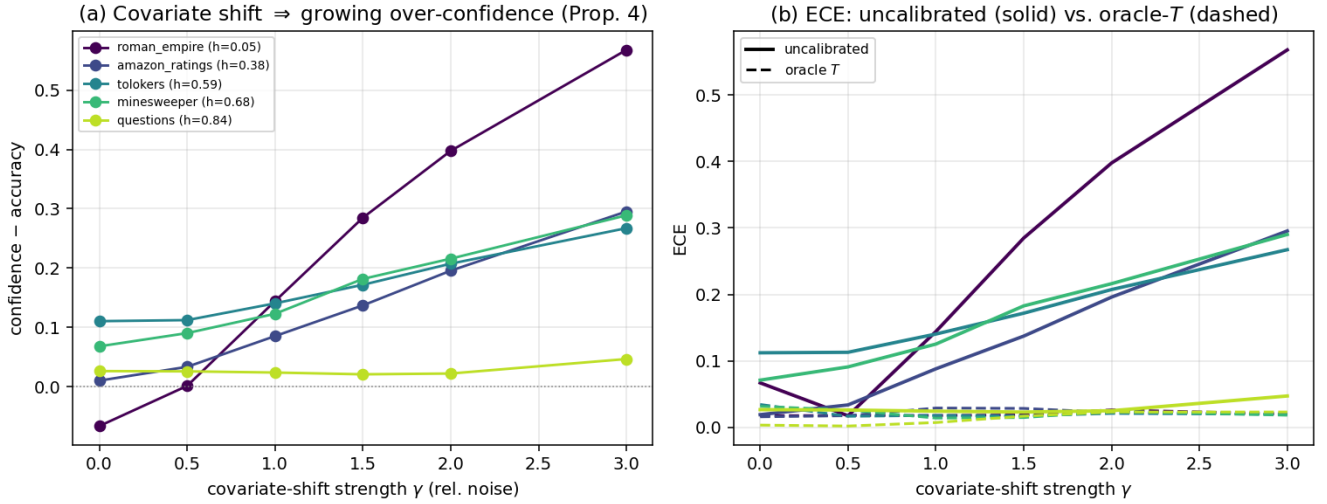


Figure 11: **Covariate-shift sweep (Proposition 3)**. (a) over-confidence grows with shift strength on every real graph; (b) uncalibrated ECE (solid) rises while a single oracle temperature (dashed) stays low.

Appendix A.1. Model and assumptions

Assumption 1 (CSBM). There are N nodes and K balanced classes with labels y_i i.i.d. uniform on $\{1, \dots, K\}$. Node features are $x_i = \mu_{y_i} + \xi_i$, with $\xi_i \sim \mathcal{N}(0, \sigma^2 I_F)$ i.i.d. and class means satisfying $\|\mu_c\| = r$ for all c and $\sum_c \mu_c = 0$ (for $K=2$, $\mu_+ = r\hat{\mu}$, $\mu_- = -r\hat{\mu}$ for a unit vector $\hat{\mu}$). Edges are drawn independently: for $i \neq j$, $\Pr(i \sim j) = p$ if $y_i = y_j$ and q otherwise. Write the edge homophily $h = p/(p + q(K-1))$ and the signal-to-noise ratio $\rho = r^2/\sigma^2$.

Assumption 2 (Large-degree regime). The expected degree $d = p(\frac{N}{K}-1) + q\frac{N(K-1)}{K}$ satisfies $d \rightarrow \infty$ as $N \rightarrow \infty$, with h and ρ held fixed.

Assumption 3 (Source-calibrated Bayes-direction classifier). The classifier is linear with weight along the Bayes-optimal direction ($\hat{\mu}$ for $K=2$; the class-mean directions for $K>2$) and zero bias, optimal for the symmetric Gaussian model of Assumption 1. Its scalar scale $a > 0$ is fixed by requiring calibration on the *source* graph, i.e. $\kappa(h_s) = 1$. A consistent estimator such as regularization-free logistic regression recovers this direction and, after source temperature scaling, this scale.

Assumption 4 (Regular-degree approximation, self-loop operator only). For Proposition 1 I approximate all degrees by the expected degree d in the symmetric normalization $\hat{A}_{ij} = ((d_i+1)(d_j+1))^{-1/2} \approx (d+1)^{-1}$. The error is $O(\text{Var}(d_i)/d^2)$ and vanishes under Assumption 2 for concentrated degrees.

Throughout, $\sigma(u) = (1 + e^{-u})^{-1}$ and φ is the standard normal density.

Appendix A.2. Proof of Lemma 1 (aggregated coordinate)

Proof. For a neighbour j , $\hat{\mu}^\top x_j = \hat{\mu}^\top \mu_{y_j} + \hat{\mu}^\top \xi_j = y_j r + \zeta_j$ with $\zeta_j \sim \mathcal{N}(0, \sigma^2)$ i.i.d. (since $\hat{\mu}^\top \mu_{\pm} = \pm r$). Hence $g_i = \frac{1}{d_i} (r \sum_{j \sim i} y_j + \sum_{j \sim i} \zeta_j)$. Given $y_i = +1$, each neighbour is same-class ($y_j = +1$) with probability h and opposite ($y_j = -1$) with probability $1-h$, independently; thus with $k = \#\{j : y_j = +1\} \sim \text{Binomial}(d_i, h)$ I have $\sum_j y_j = 2k - d_i$, so $\mathbb{E}[\sum_j y_j] = d_i(2h-1)$ and $\text{Var}[\sum_j y_j] = 4\text{Var}(k) = 4d_i h(1-h)$. The noise sum has mean 0 and variance $d_i \sigma^2$ and is independent of the labels. Dividing by d_i gives the stated mean and variance exactly. Asymptotic normality follows from the Lindeberg CLT applied to the independent bounded label terms and the Gaussian noise term. $\square$

Remark 8 (Finite-degree error). By the Berry-Esseen theorem the CDF of g_i differs from its Gaussian limit by $O(d_i^{-1/2})$. Empirically the closed forms hold to a few percent already at $d \approx 20$ (Section 6); accuracy degrades only when target accuracy approaches $\frac{1}{2}$ or 1, where the calibration slope is inherently ill-conditioned.

Appendix A.3. Proof of Theorem 1 and Corollary 1

Proof. Take $t > 0$ (the case $t < 0$ is symmetric), so the prediction is $+1$ and the event “correct” is $\{y = +1\}$. With balanced classes and the two Gaussian likelihoods,

$$\Pr(y=+1 \mid \delta=t) = \frac{\varphi\left(\frac{t-m}{s}\right)}{\varphi\left(\frac{t-m}{s}\right) + \varphi\left(\frac{t+m}{s}\right)} = \frac{1}{1 + \exp\left(-\frac{(t+m)^2 - (t-m)^2}{2s^2}\right)} = \frac{1}{1 + e^{-2mt/s^2}} = \sigma(\kappa t).$$

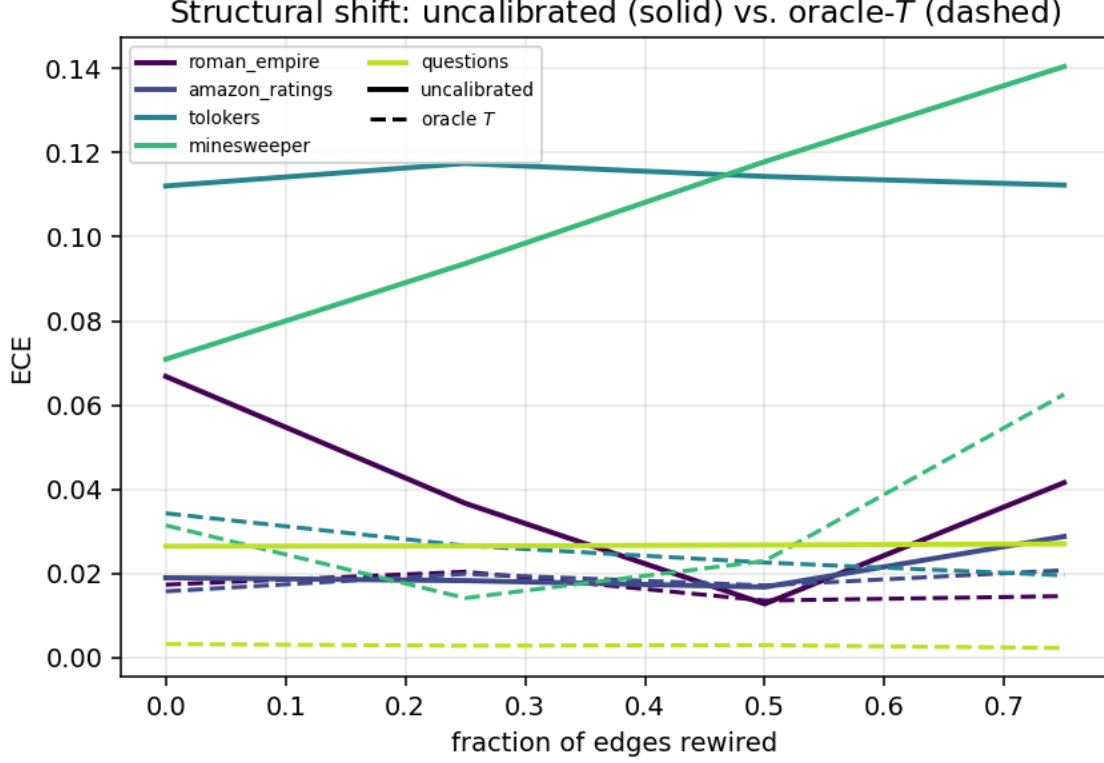


Figure 12: **Structural shift.** Uncalibrated ECE (solid) vs. oracle temperature (dashed) as target edges are rewired; one temperature recovers calibration across graphs.

The model outputs $\sigma(\delta)$ as the probability of class +1; for a prediction at $\delta = t > 0$ its confidence is $\sigma(t)$. Calibration requires $\sigma(t) = \sigma(\kappa t)$ for all t , i.e. $\kappa = 1$; the sign of $\kappa - 1$ gives over-/under-confidence. Applying temperature T maps confidence to $\sigma(t/T)$, which equals the true posterior $\sigma(\kappa t)$ for all t iff $1/T = \kappa$. Because the tempered model then coincides with the true conditional distribution, it minimizes the population NLL; hence $T^* = 1/\kappa$. Finally,

$$\kappa = \frac{2m}{s^2} = \frac{2a(2h-1)r}{a^2(4h(1-h)r^2 + \sigma^2)/d} = \frac{2d(2h-1)r}{a(4h(1-h)r^2 + \sigma^2)}.$$

Source calibration $\kappa(h_s) = 1$ gives $a = \frac{2d(2h_s-1)r}{4h_s(1-h_s)r^2 + \sigma^2}$; substituting into $\kappa(h_t)$, the common factors $2dr$ cancel, and dividing numerator and denominator by σ^2 (so $r^2/\sigma^2 = \rho$) yields Eq. (1). Degree cancels.

For Corollary 1: under homogeneous shift, $T^* = 1/\kappa$ makes the model equal to the true conditional, achieving the minimum of population NLL and zero population calibration error, by the argument above applied pointwise. Population NLL as a function of a per-node scaling T_i is, at fixed δ_i , strictly convex in $1/T_i$ with a unique minimizer $1/T^*$; therefore any assignment $\{T_i\}$ with $T_i \neq T^*$ on a positive-measure set strictly increases population NLL relative to the constant assignment $T_i \equiv T^*$. Thus per-node temperatures cannot improve calibration under a homogeneous κ ; they can only help when κ varies across nodes (heterogeneous shift). $\square$

Appendix A.4. Proof of Proposition 1 (self-loop GCN)

Proof. Under Assumption 4, $z_i = \frac{1}{d+1}(x_i + \sum_{j \sim i} x_j)$. Projecting on $\hat{\mu}$ and conditioning on $y_i = +1$: the self term contributes $r + \zeta_i$ (same-class, deterministic mean $+r$); the d neighbour terms contribute as in Lemma 1. Hence $\mathbb{E}[\hat{\mu}^\top z_i] = \frac{r}{d+1}(1 + d(2h-1)) = \frac{rA(h)}{d+1}$, and $\text{Var}[\hat{\mu}^\top z_i] = \frac{1}{(d+1)^2}(4dr^2h(1-h) + (d+1)\sigma^2) = \frac{\sigma^2 B(h)}{(d+1)^2}$ (the self term adds one unit of noise variance but no label variance). With $\delta = a \hat{\mu}^\top z_i$ I get $\kappa_{SL}(h) = \frac{2m}{s^2} = \frac{2arA(h)/(d+1)}{a^2\sigma^2 B(h)/(d+1)^2} = \frac{2r(d+1)A(h)}{a\sigma^2 B(h)}$. Fixing a by $\kappa_{SL}(h_s) = 1$ and forming the ratio, all factors except A and B cancel, giving the claim. As $d \rightarrow \infty$, $A(h) \sim d(2h-1)$ and $B(h) \sim d(4h(1-h)\rho+1)$, recovering Theorem 1. $\square$

Appendix A.5. Proof of Proposition 2 (K classes)

Proof. A neighbour is same-class (probability h), contributing μ_c , or different-class (probability $1-h$) drawn uniformly among the $K-1$ other classes, with conditional mean $\frac{1}{K-1} \sum_{c' \neq c} \mu_{c'} = -\frac{\mu_c}{K-1}$ because $\sum_{c'} \mu_{c'} = 0$. Hence

$$\mathbb{E}[z_i | y_i=c] = h\mu_c + (1-h)\left(-\frac{\mu_c}{K-1}\right) = \mu_c \frac{h(K-1) - (1-h)}{K-1} = \mu_c \frac{hK-1}{K-1}.$$

Since this scaling is common to all class directions, the source-trained logits scale by $c_K(h_t)/c_K(h_s)$ under shift. $\square$

Remark 9. The exact multiclass calibration slope requires the order statistics of the softmax and is not needed here: Corollary 1’s mechanism (uniform logit rescaling $\Rightarrow$ a single temperature suffices) carries over, and is confirmed numerically for $K=3, 4, 5$ (Fig. 6).

Appendix A.6. Proof of Proposition 3 (covariate shift)

Proof. Extra zero-mean feature noise leaves m unchanged and increases the logit variance to $s^2(\tau) = a^2(4h(1-h)r^2 + \sigma^2 + \tau^2)/d$. Thus $\kappa_{\text{cov}} = 2m/s^2(\tau) = \kappa(0) \cdot s^2(0)/s^2(\tau)$; with $\kappa(0) = 1$ (source calibration) and dividing by σ^2 , the formula follows, and $T^* = 1/\kappa_{\text{cov}}$. $\square$

Appendix A.7. Proof of Proposition 4 (ECE bound)

Proof. Population ECE = $\mathbb{E}_\delta[|\Pr(\text{correct} | \delta) - \text{conf}(\delta)|]$. By Theorem 1, at $\delta = t$ this integrand is $|\sigma(\kappa t) - \sigma(t)|$ (with $t = |\delta|$ the predicted confidence’s logit). Since σ is $\frac{1}{4}$ -Lipschitz, $|\sigma(\kappa t) - \sigma(t)| \leq \frac{1}{4}|\kappa t - t| = \frac{1}{4}|\kappa - 1||t|$. Taking expectation over δ gives the bound. It vanishes as $\kappa \rightarrow 1$ (no shift) and grows with the logit magnitude. $\square$

Appendix A.8. Discussion of scope

The results are exact in the large-degree Gaussian limit (Assumption 2) with a Bayes-direction, source-calibrated linear classifier (Assumption 3). Two honest caveats, expanded operationally in Section 7.5: (i) finite-degree corrections are $O(d^{-1/2})$ (Remark 8), negligible in the moderate-accuracy regime but growing as accuracy approaches $\frac{1}{2}$ or 1; (ii) for deep nonlinear GNNs the closed forms hold only qualitatively, though the operational consequence (a single global temperature suffices under homogeneous shift) is confirmed on real graphs in Section 6.4 (e.g. amazon-ratings ECE $0.085 \rightarrow 0.024$ after one oracle temperature).

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